

25 Years of PON standardization

Tracking the Evolution of Technology and its Market

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Abstract *Passive optical networks have a long history of standardization that has tracked both the progress of the technology and the development of the broadband market. The goal is to find a good cost-performance trade-off. This paper reviews the major themes of this work. ©2026 The Author(s)*

Introduction

Once practical low-loss optical fibre was developed, it did not take long before researcher imagined its use in access networks. At first, it was not clear which topology would be selected. Possible contenders included home-run fibre (recreating the copper network with glass), active double star (a fiberized digital loop carrier system), and passive double star - what is now known as passive optical networks (PON). In the end, PON has become the most widely deployed fibre access system, accounting for over 1 billion homes served over the entire world [1].

The initial motivations behind PON were to reduce the amount of fibre and the number of optoelectronics needed to reach a set of customers. In the 1980's, this made good sense. Over time, both fibre and transceiver modules have declined in cost; however, the cost of the labour to deploy, connect, and maintain fibres and modules has risen. At the same time, PON has built its own ecosystem, and volume cost reductions have made it very hard to beat. Most importantly, because the optical distribution networks (ODNs) of PON have been deployed so widely, they are the de-facto optical access network. No operator would want to replace their ODNs with something else, and so the network has become firmly set on a PON pathway.

Standards have led the development of PON. Other technologies were initiated with proprietary solutions that were later covered by standards. In contrast, PONs were the subject of ITU-T standards even before they were commercially relevant. This work was done in Question 2 of Study Group 15 (Q2/15) [2]. The standards closely followed the results of academic and industrial research, and they represent the cutting edge of cost-effective optical transmission technology.

The PON story is now over 25 years long, full of advancements, setbacks, and surprises. Through all of it, several major themes can be seen. The remainder of this paper reviews these: capacity, complexity, interoperability, coexistence, and support for wireless. Lastly, we consider what the next major trend in PON might be.

Optical capacity

Passive Optical Network (PON) systems have experienced rapid increases in speed over the past three decades, driven by rising demand for high-bandwidth applications (see Fig. 1). Early deployments like A-PON/B-PON in the 1990s offered speeds around 155–622 Mbps, sufficient for basic internet and voice services at the time [3]. However, as broadband usage expanded, these systems were quickly surpassed by G-PON, standardized in the early 2000s, delivering up to 2.5 Gbps downstream and 1.25 Gbps upstream [4]. GPON became widely adopted due to its efficiency and scalability.

The next major leap came with XG- and XGS-PON, offering 10 Gbps [5-6]. These technologies were deployed in the 2010's, and addressed the growing need for higher upload capacity, especially with the rise of cloud storage, video conferencing, and user-generated content. There was a major development of time and wavelength division multiplex (TWDM) PON [7]. This is a feature rich solution that has not seen much deployment.

More recently, 50G-PON has begun deployment, pushing the boundaries even further [8]. These systems are designed to support next-generation applications while maintaining compatibility with existing fibre infrastructures. Looking ahead, researchers are focusing on very high speed PON (VHSP), which will deliver speeds of up to 200 Gb/s [9]. Overall, the PON industry has increased system capacity by orders of magnitude over this time.

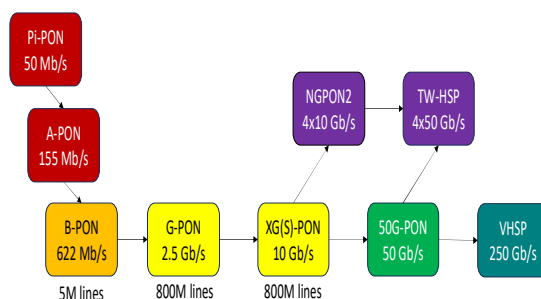


Fig. 1: The progression of PON capacities.

Processing Complexity

One of the key design aspects of the PON system is the media access control (MAC) protocol, which managed the communications between the central optical line terminal (OLT) and the user optical network unit (ONU). When PON was new, this would often be implemented using field programmable gate arrays (FPGA), and as a result the complexity of the protocol was quite limited. A-PON only implemented the bare minimum of time division multiple access (TDMA) and physical layer operations administration and management (PLOAM) functions. This was sufficient to build the first proof of concept systems, but it was clear that improvements were needed.

B-PON implemented dynamic bandwidth allocation (DBA), which allows the OLT to learn about the traffic demands of each ONU and the arbitrate the division of the limited PON upstream to those demands [10]. The concept of the transmission container (T-Cont) was introduced to limit the number of buffers in the MAC device.

As the world moved toward a packet-based network, the PON standards followed. The G-PON MAC introduced generic encapsulation mode (GEM) to carry payload packets across the PON [11]. The GEM header provides both destination tagging (the port ID) as well as packet fragment length information. This data sub-layer enables packets to be fragmented over multiple transmission opportunities. This increases the transport efficiency. In general, all the elements of the GEM protocol were squeezed to the minimum size, due to the fact that the line rate was still relatively small (1.2 Gb/s in the upstream). G-PON also included a modern cipher algorithm (AES), far better than the previous algorithm.

When it came time for XG-PON, the PON industry had become a well-established ecosystem, and the limitations of FPGAs largely went away. The biggest complexity increase in XG-PON was the mandatory inclusion of forward error correction (FEC). The Reed-Solomon algorithm was used, which delivered a needed 3~4 dB sensitivity improvement. XG-PON also reformed the data plane protocol to XGEM, which justified all the data elements to 4 byte boundaries, and made the header length 8 bytes to make space for improved functions.

50G-PON continued this trend, changing its FEC algorithm to the low density parity check (LDPC) type. Some implementations also have digital signal processing (DSP) equalization functions [12]. The later generations of MAC are far more complex, but their cost has not increased due to the improvements of silicon microelectronics. In fact, many devices implement multiple PON flavours to save development costs.

Interoperability

The history of interoperability in PON systems reflects the broader evolution of fibre access from proprietary deployments to highly standardized, multi-vendor ecosystems. In the earliest stages of A-PON and B-PON interoperability was limited. Although standards existed such as the ONU management and control interface (OMCI), these standards were written in an open-ended way, and they allowed multiple options for how to set up the data networking in the PON equipment [13]. As a result, vendors often implemented features differently, resulting in systems where ONUs and OLTs from different manufacturers were not easily interchangeable. As a result, network operators were typically locked into single-vendor solutions.

The interoperability issue was addressed starting with the B-PON system and continuing with G-PON. Industry groups such as the Full Service Access Network (FSAN) played a key role in aligning vendors and defining common specifications [14]. A lot of work was done to make tough decisions on questions like “how many T-Conts are needed?” and “how are multiple Ethernet ports networked?” In addition to the theoretical design work, real life interoperability testing events, often called “plug-fests” became common, allowing vendors to validate cross-compatibility in controlled environments. These tests reveal implementation differences, which would then feed back to the standards development.

To take interop to the next level, more formal processes were established in the Broadband Forum [15]. Several technical requirements documents (e.g., TR-156 and TR-167) were developed to itemize the many features and aspects that interoperable G-PON equipment must support [16]. Building on these, a detailed test plan for ONU conformance was specified. This plan was used to establish a testing program that was managed by an independent testing laboratory. At present many ONUs have been certified as conformant to the standards.

From G-PON, all the progress on OMCI and interoperability has been carried over from one system to the next. This happens naturally, because all systems use the common OMCI. As each new generation of PON has emerged, interoperability has been achieved quicker and easier. While challenges still exist—particularly around advanced features and performance optimization—the industry has largely succeeded in creating interoperable solutions.

Coexistence

The concept of coexistence in PON systems has been central to design of access networks. Early

PON deployments such as A-PON and B-PON did not prioritize coexistence; upgrading required overlay networks or full replacement of equipment. As PON investment grew, operators demanded ways to introduce new technologies without disrupting existing.

Coexistence began with G-PON, which introduced wavelength planning that limited the transmitter wavelength bands and protected the receivers with blocking filters. This prior planning enabled the XG-PON and XGS-PON systems to be designed with wavelength bands that were separate from G-PON [17]. As a result, higher-speed systems can operate alongside G-PON on the same ODN. This approach allowed operators to upgrade select users to faster services without replacing the entire network. The initial architecture used passive wavelength division coexistence elements (CEs) that connecting multiple OLT ports to a common ODN. CEs enable the existing OLT to remain when reinforcing with the next generation. For higher density, the architecture of multi-PON modules (MPM) was introduced, where each MPM contains the optics for both system's OLT plus the coexistence element.

When 50G-PON was developed, its coexistence with previous generations spurred many considerations [18]. The initial idea was to support coexistence with either G-PON or XG(S)-PON, but not both. As time went on, both the technology and the market shifted, and so "triple coexistence" was standardized, allowing G-, XG(S)-, and 50G-PON all on the same ODN. Triple MPMs are now available, which is quite a feat of opto-mechanical design [19].

Looking forward, VHSP systems will need to coexist with some selection of prior PON systems. Due to the low chromatic dispersion of the central O-band, some VHSP systems would be easier if G-PON was retired before the VHSP upgrade. However, history has shown that access systems have a long life, and it might be a long time before G-PON is fully gone. Thus, research is exploring how to enable four-way coexistence by using advanced modulation techniques [20].

Support for Wireless

The primary application space for PON is residential services, which is justified by the huge number of residential users. This drives the volume of PON equipment, and reduces its cost. Once optical access became ubiquitous, it was naturally extended to similar applications like multiple dwelling units (MDUs) small and medium enterprises (SMEs). Such extensions are straightforward, differing from residential only in the bandwidth and interfaces supported.

In contrast, serving radio access networks comes with some special requirements that are

not automatically satisfied with a residential PON system. PON-fed RAN is a very attractive business due to the wide adoption of wireless service, and so work was initiated to find a solution.

Modern RANs are quite complex, and there are multiple generations of wireless system, and then multiple styles of radio network architecture. Q2/15 developed a supplement that analysed the entire landscape of wireless [21]. This considered all the functional split options and developed an inventory of basic requirements for the optical access system.

Most systems require the delivery of very precise (~microsecond accuracy) time of day. In most parts of the network, this is accomplished by using IEEE 1588 precision time protocol (PTP). However, PTP doesn't work over PON because the delay in the two directions is not the same, which violates the PTP underlying assumption. To address this, the OLT convey to the ONU the precise time of day over a PON specific channel [22]. Then the ONU can determine the time of day.

Lower layer split systems require very low latency. PON presents some issues in this regard. The ranging process inevitably causes an outage of about 250 microseconds from time to time. This can be avoided if the ranging is accomplished on a separate PON system. The DBA process adds delay because it is essentially a three-way handshake between the OLT and ONU. Cooperative DBA can reduce this by coupling the OLT scheduler to the wireless controllers, so that the OLT can predict when each ONU will need bandwidth [23]. Ultimately, the most extreme front-haul wireless systems need dedicated bandwidth channels in the access. For this, wavelength division multiplexed (WDM)-PON is a strong candidate [24]. Several WDM-PON systems have been standardized; however, they remain suitable for only niche applications.

Conclusions

The development of PON has had a huge impact on the access network, and the whole network. Beyond the themes described above, the next big frontier for PON seems to be artificial intelligence (AI). So far, the access network has been all about transport and delivery of content to users. The recent feasibility of AI now brings us three opportunities. AI could be used to perform various tasks for the broadband network. Alternatively, the network could provide support for AI functions. Then, by combining the unique capabilities of broadband and AI, altogether new services could become possible. These are being explored now [25], and they could become the defining features of the next PON systems that move into deployment.

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